

EE2 Circuits and Systems

Problem Class 1 (Sheet 1)

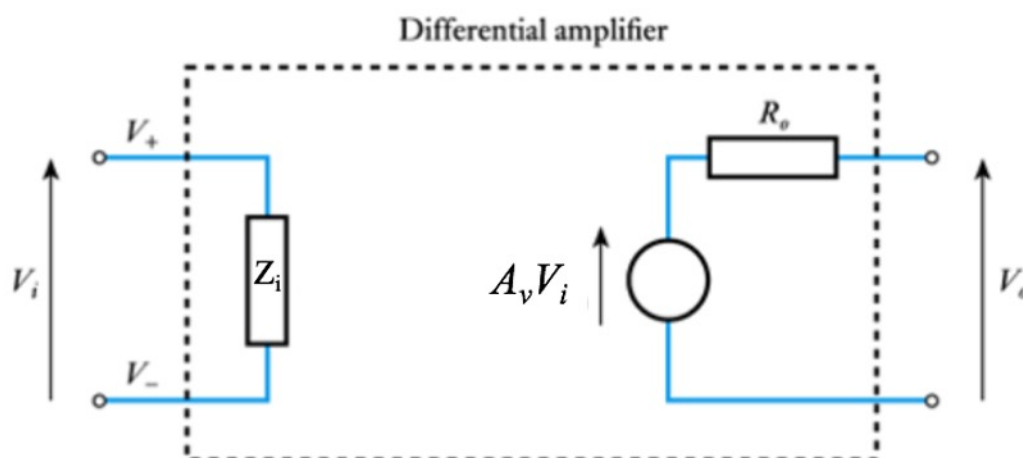
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Test Yourself - Problem 1 (Sheet 1 Q1a)

The MCP601 op-amp is modelled using the simplified circuit model shown below. Find the corresponding parameters from the datasheet where appropriate.

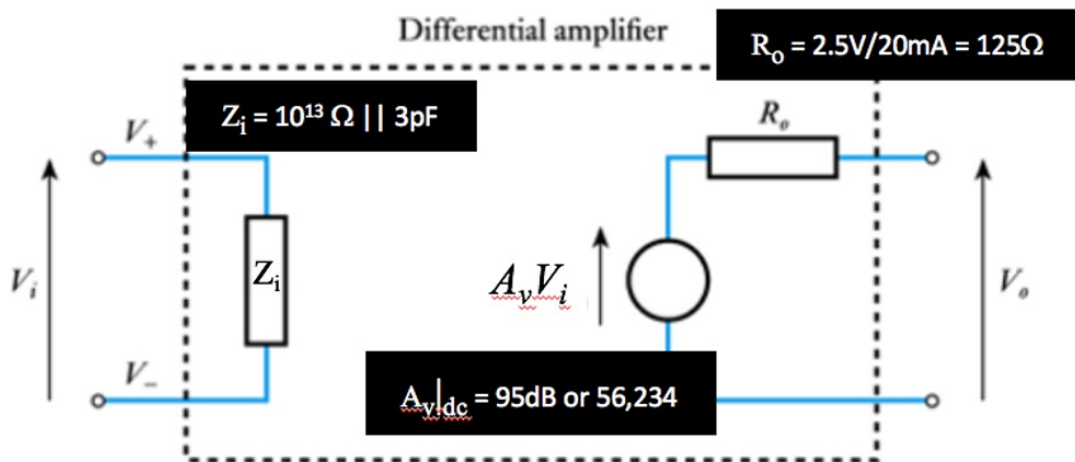
(MCP601 datasheet provided)



Test Yourself - Solution 1 (Sheet 1 Q1a)

Common Mode Input Impedance	Z_{CM}	—	$10^{13} 6$	—	ΩpF
Differential Input Impedance	Z_{DIFF}	—	$10^{13} 3$	—	ΩpF
DC Open Loop Gain	A_{OL}	95	110	—	dB
Output Short Circuit Current	I_{SC}		20	—	mA

$R_L = 5k\Omega$ to $V_{DD}/2$,
 $100mV < V_{OUT} < (V_{DD} - 100mV)$
 $V_{OUT} = 2.5V$,
 $V_{DD} = 5V$



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Test Yourself - Problem 2 (Sheet 1 Q1b, c)

- b) What voltage range at the output?
- c) You are required to build a voltage amplifier for a signal in the frequency range of 10Hz to 100kHz, what is the maximum gain you can expect to achieve with one of these op-amps?

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Test Yourself - Solution 2 (Sheet 1 Q1b)

b) What voltage range at the output?

Linear Region Maximum Output Voltage Swing

V_{OUT}	$V_{SS} + 0.050$	—	$V_{DD} - 0.050$	V	$R_L = 25k\Omega$ to $V_{DD}/2$, $A_{OL} \geq 100dB$
V_{OUT}	$V_{SS} + 0.100$	—	$V_{DD} - 0.100$	V	$R_L = 5k\Omega$ to $V_{DD}/2$, $A_{OL} \geq 95dB$

- 1) If the output load is high (25k) i.e. output current is low, output range can be $\pm 50mV$ from the power supply voltages
- 2) This output range is dependent on the output current. As R_L is reduced to 5k, the output range is reduced because the so-call headroom is now increased to $\pm 100mV$.

Test Yourself - Solution 2 (Sheet 1 Q1b)

b) What voltage range at the output?

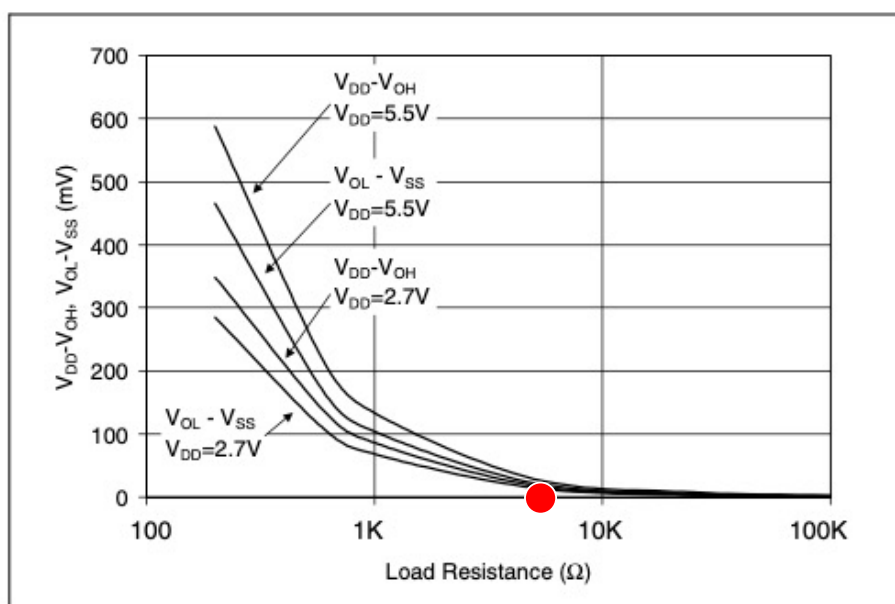
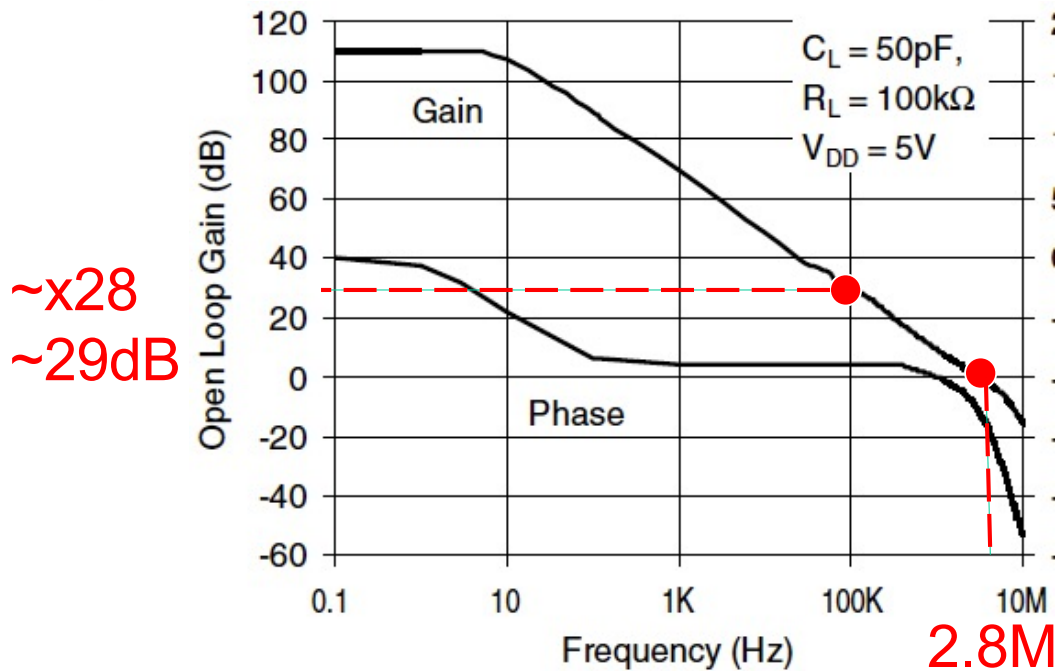


FIGURE 2-20: Low Level and High Level Output Swing vs. Resistive Load

Test Yourself - Solution 2 (Sheet 1 Q1b)

- c) You are required to build a voltage amplifier for a signal in the frequency range of 10Hz to 100kHz, what is the maximum gain you can expect to achieve with one of these op-amps?



Test Yourself - Problem 3 (Sheet 1 Q1d)

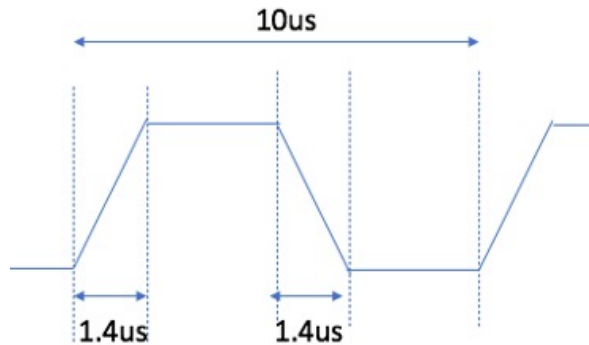
- d) This op-amp is used as a unity gain buffer amplifier (voltage follower) circuit. The input is driven directly with a 100kHz a digital clock signal between 0V and 3.3V. Sketch the expect output of the buffer amplifier.

Test Yourself - Solution 3 (Sheet 1 Q1d)

- d) This op-amp is used as a unity gain buffer amplifier (voltage follower) circuit. The input is driven directly with a 100kHz a digital clock signal between 0V and 3.3V. Sketch the expect output of the buffer amplifier.

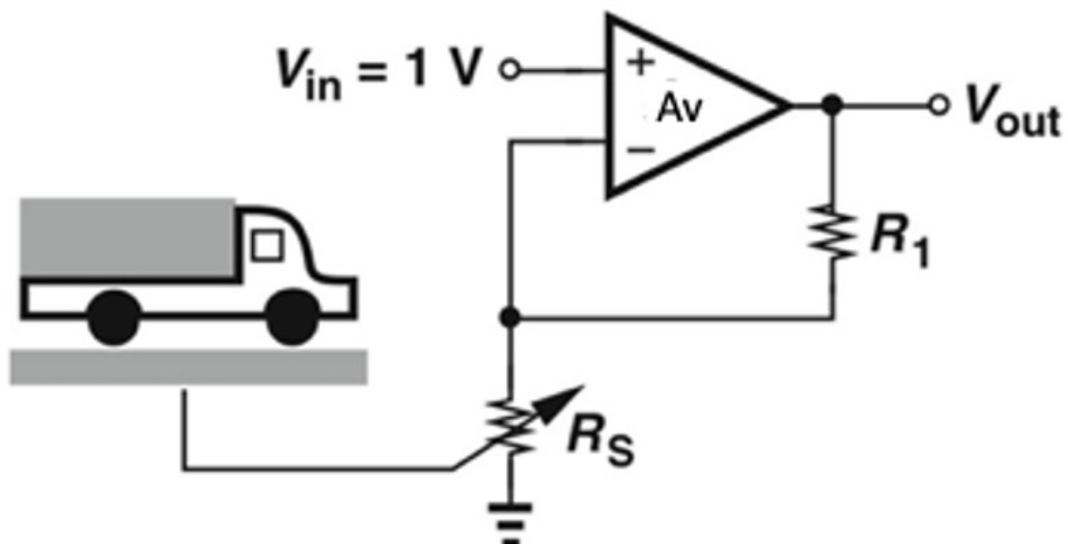
Slew Rate | SR | — | 2.3 | — | V/ μ s | $G = +1V/V$, $V_{DD} = 5V$

Therefore this typically takes $3.3/2.3 \mu s = 1.4 \mu s$ to go from 0V to 3.3V or 3.3V to 0V. Hence the waveform would look like:

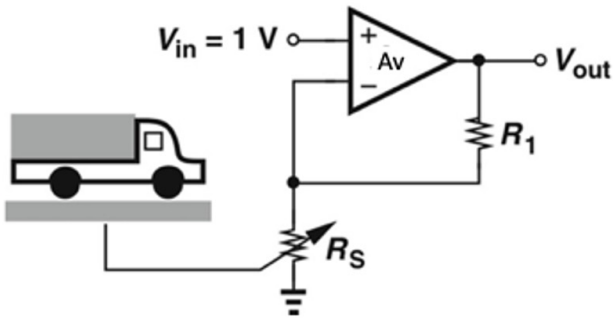


Explain it - Problem 4 (Sheet 1 Q2)

A truck weighing station incorporates a sensor whose resistance varies linearly with the weight: $R_S = R_0 + \alpha W$. Here R_0 is a constant value, α a proportionality factor and W the weight of each truck. R_S is part of an amplifier circuit shown below. Given that $V_{in} = 1V$, determine the gain of the system, defined as the change in V_{out} (i.e. δV_{out}) divided by the change in weight W (i.e. δW). You may assume that the op-amp is ideal.



Explain it - Problem 4 (Sheet 1 Q2)



The gain of the system from V_{in} to V_{out} is:

$$\frac{V_{out}}{V_{in}} = 1 + \frac{R_1}{R_S}$$

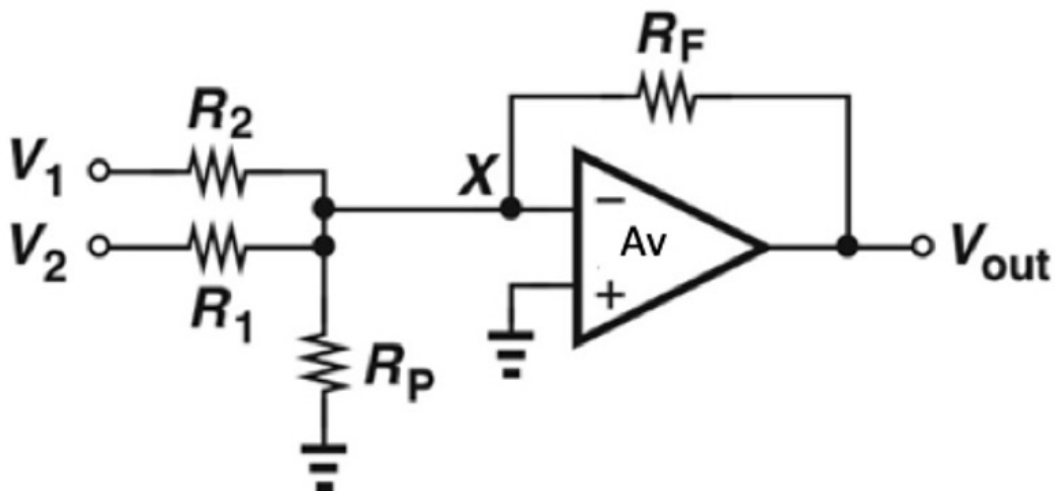
Since $V_{in} = 1$,

$$V_{out} = 1 + \frac{R_1}{R_0 + \alpha W}$$

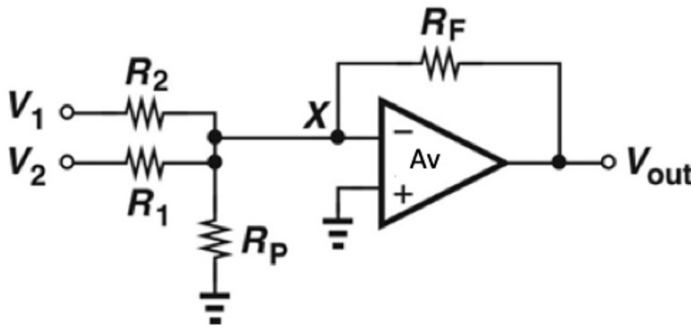
$$\frac{dV_{out}}{dW} = -R_1 \alpha / (R_0 + \alpha W)^2$$

Test yourself - Problem 5 (Sheet 1 Q3)

Due to a manufacturing error, a parasitic resistance R_P has appeared in the circuit below. Calculate V_{out} in terms of V_1 and V_2 for $A_V = \infty$ and $A_V < \infty$. (Note that R_P can also represent the input impedance of the op-amp). State any other assumptions used.



Test yourself - Solution 5 (Sheet 1 Q3)



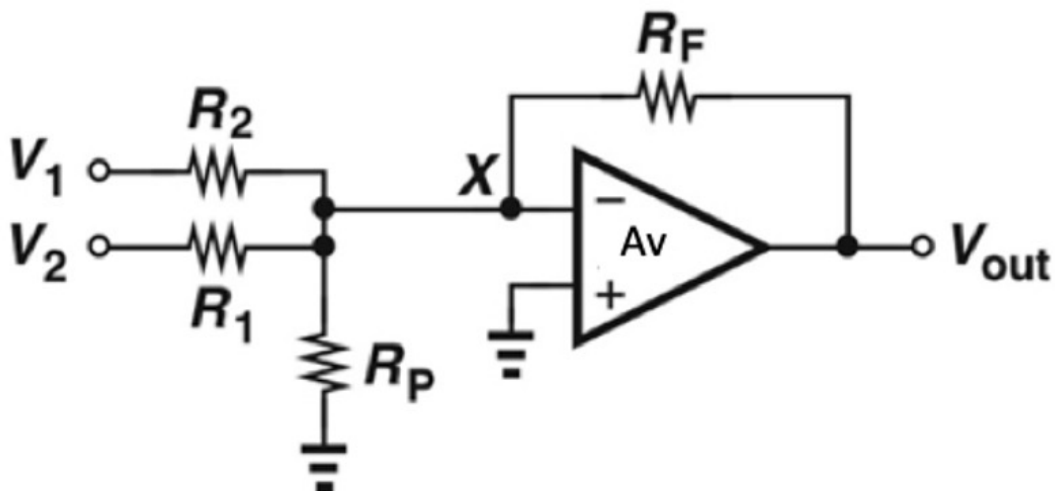
For $A_V = \infty$, $V_+ = V_-$.

Since X is the virtual earth point ($V_X = 0$), no current flows in R_P .
 V_{out} is therefore not affected by R_P at all.

$$V_{out} = -R_f \left(\frac{V_1}{R_2} + \frac{V_2}{R_1} \right).$$

Explain it - Problem 6 (Sheet 1 Q3)

Due to a manufacturing error, a parasitic resistance R_P has appeared in the circuit below. Calculate V_{out} in terms of V_1 and V_2 for $A_V = \infty$ and $A_V < \infty$. (Note that R_P can also represent the input impedance of the op-amp). State any other assumptions used.

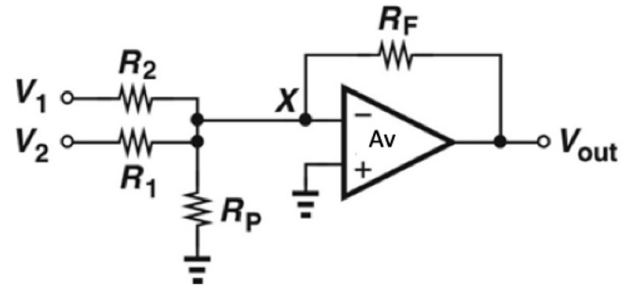


Explain it - Solution 6 (Sheet 1 Q3)

For $A_V < \infty$, we need to include the effects of R_p in the calculation.

Apply KCL at V_x :

$$\frac{V_x}{R_p} = \left(\frac{V_1 - V_x}{R_2} + \frac{V_2 - V_x}{R_1} + \frac{V_{out} - V_x}{R_f} \right)$$



We also know that: $V_{out} = -A_V V_x$

$$V_{out} = -\left(\frac{V_1}{R_2} + \frac{V_2}{R_1} \right) [R_f || A_V (R_1 || R_2 || R_p || R_f)]$$

If $R_p \gg$ all other resistances

$$[R_f || A_V (R_1 || R_2 || R_p || R_f)] \approx R_f$$